Adaptive Receptive Field Enhancement Network Based on Attention Mechanism for Detecting the Small Target in the Aerial Image

Jian Wang, *Member, IEEE*, Xinqi Li, Lihui Zhou, Jiafu Chen, Zihao He, Linyang Guo, and Jie Liu

***Abstract*— To address the problem of insufficient semantic feature information caused by small objects in aerial images, an adaptive receptive field enhancement network based on attention mechanism suitable for aerial scene is proposed. First, to make the model more suitable for deployment on unmanned aerial vehicle (UAV) platforms with limited resources, the receptive field block (RFB) is reconstructed by a series branch with feature multiplexing, which can make the output feature map of the front branch pass through the convolutional layer of the back branch for further feature extraction, improving the utilization of convolutional resources. Second, to expand the receptive field with as little loss of local contextual information as possible, Kronecker convolution (K-conv) is employed in the RFB branch for feature extraction, which can augment the original image covered by a single pixel on the feature map, expand its global semantic information and local contextual information, and improve the integrity of object extraction for small targets. Finally, to solve the problem of a fixed receptive field size of neurons at each layer in the network caused by direct aggregation feature maps, the selective convolutional module based on attention mechanism is added to the network in this article, so that neurons at each layer in the network can adaptively adjust the receptive field size, so as to make the different sizes of the detector head to play the best detection ability. In this article, experiments are carried out on a homemade aerial photography small target dataset to verify the effectiveness of the adaptive field enhancement network. The experimental results show that the adaptive field enhancement network proposed in this article can effectively improve the detection accuracy of the algorithm on targets with low resolution and insufficient semantic feature information in aerial images and is more suitable for use in aerial scene.**

***Index Terms*— Adaptive receptive field, aerial image, selective convolution, small target.**

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Jian Wang is with the Electronic and Information College, Northwestern Polytechnical University, Xi'an 710129, China, and also with the No. 365 Institute, Northwestern Polytechnical University, Xi'an 710065, China (e-mail: jianwang@nwpu.edu.cn).

Xinqi Li, Lihui Zhou, Jiafu Chen, Zihao He, Linyang Guo, and Jie Liu are with the Electronic and Information College, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: lixinqi@mail.nwpu.edu.cn; zhoulihui@mail.nwpu.edu.cn; chenjiafu@mail.nwpu.edu.cn; hezihao@mail.nwpu.edu.cn; guolinyang@mail.nwpu.edu.cn; liujie2020@mail.nwpu.edu.cn).

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I. Introduction

OBJECT detection is a fundamental visual task that involves classifying and localizing targets. As a subfield of generic object detection, small object detection focuses on detecting targets with small sizes in images, which is of great significance in the fields of autonomous driving, intelligent security, and unmanned aerial vehicle (UAV) surveillance.

The direct use of generic object detector to detect small targets will be missed and misdetected; the reason is that the existing object detection networks, a convolutional neural network, with numerous layers are usually utilized to extract high-level semantic features of images. However, as the number of layers increases, the resolution of feature maps decreases, which directly causes the blurring or even disappearance of semantic feature information of small and medium targets in feature maps. Because there are many small objects in aerial images, improving the detection performance for small objects can enhance the detection accuracy of aerial images. Aiming at the problem of small objects in aerial images, scholars improve it from several angles, and some use image cropping to enlarge the small targets in the image (see Yang et al. [1], Li et al. [2], Deng et al. [3], Huang et al. [4]); some utilize Transformer to model long-distance dependencies and capture extensive contextual information related to small targets (see Xue et al. [5], Cao et al. [6], Xu et al. [7]), and some solve the problem of insufficient semantic information of small objects by enhancing the target features.

Feature enhancement of the target includes feature fusion and receptive field enhancement. Regarding feature fusion, Pan et al. [8] proposed a multiscale feature fusion scheme to improve small object detection on the single-shot multibox detector (SSD) model. Yin et al. [9] improved SSD detection performance via feature fusion and dilated convolution. Guo and Deng [10] enhanced feature fusion by using pooling and deconvolution and the number of feature layers. Zhang et al. [11] proposed cascade fusion network (CFNet), which inserts feature integration operations into the backbone and greatly increases the richness of feature fusion. Compared with feature fusion, receptive field enhancement is a common way to achieve feature enhancement, as the receptive field is an important attribute of convolutional neural networks, and the size of the receptive field of the convolutional layer is related to the integrity and robustness of feature extraction from targets, especially for small targets. The DetNet method was

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proposed by Li et al. [12] and the receptive field enhancement (RFE) module was proposed by Yu et al. [13] in you only look once (YOLO)-FaceV2, which used dilated convolution to enhance the receptive field of the feature maps, but the sparsely sampled features of the dilated convolution result in a lack of local contextual information. Liu and Huang [14] proposed RFBNet [receptive field block (RFB)]. Based on the SSD algorithm, an RFB structure composed of parallel branches is added to enhance the feature map receptive field. However, the RFB structure obtains a variety of scales by continuously adding branches, which causes a large increase in the calculation amount of the receptive field enhancement module. Xie et al. [15] constructed a detection subnetwork with dynamic receptive fields by adding a short connection path and global contextual upsampling module and dynamically selected appropriate scale receptive field features. However, directly aggregated feature maps led to the same receptive field scale of neurons in each layer.

To solve the problem of insufficient semantic feature information caused by small targets in aerial images and as the current general object detection algorithm based on receptive field enhancement is still inadequate, this article proposes an adaptive receptive field enhancement network based on attention mechanism. First, to solve the problem that the parallel branch cannot extract features from the feature map many times, an RFB is reconstructed by a series branch with feature multiplexing, which can make the output feature map of the front branch pass through the convolutional layer of the back branch for further feature extraction. Second, to enlarge the receptive field and solve the problem of missing contextual semantic information caused by the enlargement of the receptive field by the dilated convolution method, Kronecker convolution (K-conv) is utilized in the RFB branch for feature extraction, which can augment the original image covered by a single pixel on the feature map, expand its global semantic information and local contextual information, and improve the integrity of object extraction for small targets. Finally, to solve the problem of fixed receptive field size of neurons at each layer in the network caused by direct aggregation feature maps, the selective convolutional module based on attention mechanism is added to the network in this article, so that neurons at each layer in the network can adapt to the receptive field size, so as to make the different sizes of the detector head to play the best detection ability. To reflect the superiority of the receptive field enhancement network proposed in this article, a variety of other receptive field enhancement networks were added based on the YOLOx [16] algorithm for comparison. Experiments show that the receptive field enhancement network proposed in this article is superior to other receptive field enhancement networks in improving the algorithm's detection accuracy of small targets in aerial images. To further verify the effectiveness and generalization of the receptive field enhancement network proposed in this article, it was added to the algorithms Faster region-based convolutional neural network (R-CNN) [17] and YOLOv5 [18]. Experimental results show that the adaptive receptive field enhancement network proposed in this article can effectively improve the detection rate of the algorithm on targets with low resolution and insufficient semantic feature information in aerial images. Finally, combining the network proposed in this article with YOLOx and comparing it with the algorithms of recent years, the experimental results show that our method has better detection performance.

In summary, the contributions of this article are as follows.

1) We design a new network module more suitable for use in aerial scene, which uses serial connectivity to reduce model parameters, captures part of the missing local contextual information while expanding the receptive field of the feature map, and adaptively adjusts the size of the receptive field of the neurons according to different feature map information.
2) We integrate effective squeeze-and-excitation (eSE) module into our network, which can effectively enhance the feature extraction capability of our network and avoid the loss of channel information.
3) We integrate our network into YOLOx and conduct experiments on a challenging homemade aerial small target dataset, which show that the proposed algorithm improves the mean average precision (mAP) and maintains a good detection speed.

II. Related Work

A. Object Detection on Aerial Images

Since it is difficult to extract features from small targets with high-density aggregation in aerial images, current research usually adopts a combination of coarse and fine detection, which first uses a coarse detector to detect large targets in the image and locate the high-density target region, then crops the region and uses the fine detector to detect it, and finally fuses the results of coarse and fine detection by NMS. For example, Yang et al. [1] proposed the ClusDet for fine detection on subregions. Li et al. [2] proposed the density-map guided object detection network (DMNet) using density maps of crowd counts to obtain subregions. Deng et al. [3] proposed the global-local self-adaptive network (GLSAN) algorithm incorporating a super-resolution network to achieve undistorted zoomed-in view on subregions. Huang et al. [4] proposed the multi-proxy detection network with unified foreground packing (UFPMPDet) algorithm to stitch the cropped subregions into a single image. Koyun et al. [19] proposed a two-stage object detection framework called "focus and detect" and used the incomplete box suppression method to overcome the truncation effect of the region searching method. Although these methods improve the accuracy of small targets on aerial images, they require multiple inference on an image, which increases the inference time to some extent.

B. Receptive Field and Receptive Field Block Structure

The receptive field is an important concept of the convolutional neural network. The area size of the input image corresponding to a point in the feature map of a certain layer in the convolutional neural network is the receptive field size of the layer. This receptive field size is calculated as follows:

$$R_n = R_{n-1} + (k_n - 1) \prod_{i=1}^{n-1} S_i \tag{1}$$

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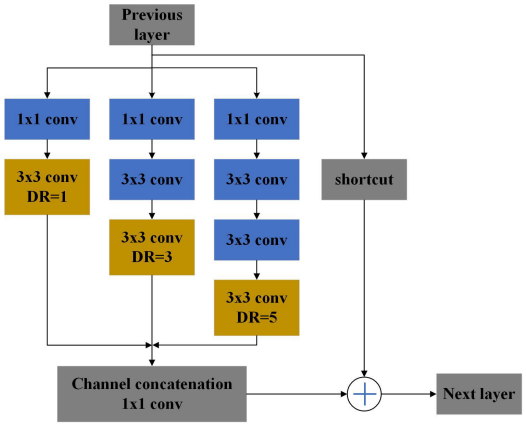


Fig. 1. RFB structure.

where R_n is the size of the receptive field of the nth layer, and k_n and S_n are the size and sliding step size of the nth layer convolution kernel, respectively.

The larger the receptive field is, the larger the scope of the original image covered by a single pixel on the feature map, the more the global semantic information is contained, the higher the semantic level of the feature, and the stronger the feature extraction ability. Therefore, the size of the receptive field of the convolutional layer is related to the integrity and robustness of the feature extraction, especially for the detection of small targets with insufficient semantic information of the feature map. When the object detection algorithm detects small targets, the stronger the feature extraction ability of the algorithm is, the more it can avoid false detection and missing detection of small targets. Therefore, it is necessary to increase the receptive field of the feature map extracted by the network.

The RFB structure is a method employed in the RFBNet [14] algorithm to expand the receptive field of feature maps. Starting from the perspective of simulating the receptive field of human vision and, thus, enhancing the feature extraction capability of the network, the RFB structure is based on the idea of inception [20]. The receptive field can be effectively increased by adding the dilated convolution to inception. The formula for calculating the receptive field using dilated convolution is as follows:

$$k_n{}' = k_n + (k_n - 1) \times (r - 1) \tag{2}$$

$$R_n = R_{n-1} + \left(k_n{}' - 1\right) \prod_{i=1}^{n-1} S_i \tag{3}$$

where k_n and $k_n{}'$ represent the actual convolutional kernel size and the equivalent convolutional kernel size of the nth layer, respectively, r is the dilation rate, and R_n is the size of the receptive field of the nth layer.

The structural diagram of the RFB is shown in Fig. 1. Dilated convolution with different dilation rates is used in the three parallel branches of RFB, expanding the receptive field at different multiplicities. Since dilated convolution is realized by zero padding, it lacks the ability to capture local contextual information. Therefore, it is necessary to conduct general convolution first to refine contextual semantic information. Since each convolution is involved in the feature output only once, this increases the number of parameters and computation of the network.

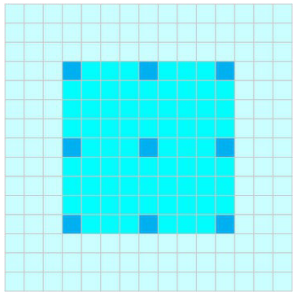


Fig. 2. Schematic of dilated convolution with a dilation rate of 4.

C. Kronecker Convolution

Dilated convolution [12] is the most commonly employed method to enhance the receptive field. Compared with ordinary convolution, dilated convolution increases the dilation rate parameter, which refers to the distance between each point in the convolution kernel. The dilation rate is often realized by zero padding, which makes it difficult for the dilated convolution to capture contextual information when the dilation rate is too large. For example, when the dilation rate is 4 in Fig. 2, only nine of the 81 vectors in the convolution block participate in the calculation.

Since the sparse sampling nature of the dilated convolution leads to a lack of contextual information, the K-conv layer [21] is introduced into the receptive field enhancement module in this article. Inspired by the Kronecker product in mathematics applications, K-conv extends the standard convolution kernel by the Kronecker product, which not only increases the receptive field of the feature map but also enables the feature map to capture local contextual information. If A is the $m \times n$ matrix, $[a_{11} \cdots a_{mn}]$ is the matrix $\boldsymbol{A}$ coefficient, $\boldsymbol{B}$ is the $r \times s$ matrix, and $\boldsymbol{A} \otimes \boldsymbol{B}$ is the $mr \times ns$ matrix, then the mathematical definition of the Kronecker product is presented as follows:

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}. \tag{4}$$

After understanding the mathematical definition of the Kronecker product, K-conv is described in detail below. For standard convolution, the input feature map is represented as $A \in R^{H_A \times W_A \times C_A}$, and the output feature map is represented as $B \in R^{H_B \times W_B \times C_B}$, where, H, W, and C represent the height, width, and number of channels, respectively, of feature map A and feature map B. C_A and C_B represent the number of channels of feature map A and feature map B, respectively. In standard convolution, the convolution kernel is represented as $K \in R^{C_A \times C_B \times (2k+1) \times (2k+1)}$, and the bias is represented as $b \in R^{C_B}$. The transformation matrix F is introduced into K-conv, and F is set as a fixed $r_1 \times r_1$ matrix. r_1 is an intermediate expansion factor that controls the convolution dilation rate. By calculating the Kronecker product of the transformation matrix F and the convolution kernel K, the convolution kernel in K-conv is expanded from $(2k + 1) \times (2k + 1)$ to $(2k+1)r_1 \times (2k+1)r_1$. To avoid introducing extra parameters,

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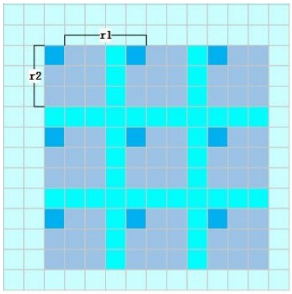


Fig. 3. Schematic of K-conv with $r_1 = 4$ and $r_2 = 3$.

matrix F is set as a combination of matrix I and zero matrix O, where I is a square matrix of $r_2 \times r_2$, and all elements are 1. The size of the subregion is controlled by setting the internal sharing factor r_2 to capture eigenvectors and shared filter vectors. Therefore, the convolution kernel K' in K-conv is expressed as follows:

$$K'(c_2, c_1) = K(c_2, c_1) \otimes F \tag{5}$$

$$F = \begin{bmatrix} I_{r_2 \times r_2} & \\ & O_{(r_1 - r_2) \times (r_1 - r_2)} \end{bmatrix} \tag{6}$$

where $c_1 \in [1, C_A]$ and $c_2 \in [1, C_B]$.

For example, the K-conv convolution of $r_1 = 4$ and $r_2 = 3$ is shown in Fig. 3, where r_1 controls the convolution dilation rate and r_2 controls the subregion size to capture the feature vector. There are 81 vectors in the convolutional block involved in the calculation in Fig. 3.

Compared with the dilated convolution that simply inserts the zero-expanded convolution kernel, K-conv realizes the expansion of the ordinary convolution kernel by performing the Kronecker product of the transformation matrix F and the convolution kernel K, which avoids the vector being calculated at zero. The intermediate expansion factor r_1 controls the dilation rate. When r_1 increases, the receptive field will also be enlarged. Since the transformation matrix F only contains 1 and 0 values, additional parameters are not introduced. In addition, the transformation matrix F has an identity submatrix of $r_2 \times r_2$, which causes more vectors to participate in the correlation calculation, and K-conv can capture the local contextual information that is disregarded due to the dilated convolution.

III. Adaptive Receptive Field Enhancement Network Design

To solve the problem of insufficient semantic feature information caused by small aerial image targets, this article proposes an adaptive receptive field enhancement network based on attention mechanism from the perspective of the receptive field after analyzing the dilated convolution and RFB structure in Sections II-B and II-C. First, to solve the problem that in the RFB structure, multiscale receptive fields are obtained by increasing the number of parallel branches and increasing the dilation rate, resulting in a large increase in the amount of calculation, it is proposed to reconstruct the RFBs by a series of branches with feature multiplexing, so that the output feature map of the front branch can be further extracted through the convolutional layer of the back branch.

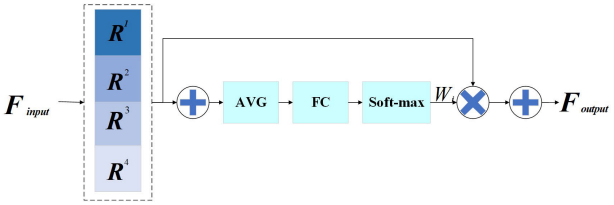


Fig. 4. Selective convolution module based on attention mechanism.

Second, to increase the receptive field and to solve the problem of missing contextual semantic information caused by the enlargement of the receptive field by the dilated convolution method, it is proposed to use K-conv in the RFB branch for feature extraction, which can augment the original image covered by a single pixel on the feature map, expand its global semantic information and local contextual information, and improve the integrity of target extraction. Finally, to solve the problem that the size of the receptive field of neurons at each layer in the network is fixed due to the direct aggregation of feature maps, it is proposed to add the selective convolutional module based on attention mechanism to the network to enable the neurons at each layer to adjust the size of the receptive field adaptively.

A. Selective Convolution Module Based on Attention Mechanism

In the biological neural field, the size of the receptive field of neurons in the visual cortex is adjusted according to the size of the stimulus, while the size of the receptive field of each layer of neurons in a standard convolutional neural network is designed to be the same size. Therefore, to realize the adaptive adjustment of the receptive field size of each layer of neurons according to the multiscale information of the input information, a selective convolution module based on attention mechanism is employed [22] to realize the aggregation of feature maps according to different weights, as shown in Fig. 4.

The input feature map $F_{\text{input}} \in R^{C \times H \times W}$, $R^i (i \in 1\text{–}4)$ represents the feature map of different scale outputs after the feature extraction branch (four branches are used as an example here, and the actual number of branches can be determined according to demand). To enable neurons to adaptively adjust the size of the receptive field adaptively according to the stimulus content, it is necessary to calculate the influence of different receptive field feature maps on the next level, namely, the weight value.

First, the feature maps of multiple branches are fused by summing

$$R = \sum_{i=1}^{4} R_i. \tag{7}$$

Second, the global information is embedded via global average pooling [23], that is, all pixels of the feature map are added and averaged, and a value is obtained to represent the corresponding feature map $s \in R^C$. Therefore, the feature map s_c of the cth channel after global average pooling is expressed

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as follows:

$$s_c = F_{\text{avg}}(R_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{i=1}^{W} R_c(i, j). \tag{8}$$

Third, the fully connected (FC) layer is used to predict the probability vector $V \in R^{C'}$ of different feature maps

$$V = F_{fc}(s) = \delta(B(W_s)) \tag{9}$$

where δ is the loss function [24], B denotes regularization [25], and $W_s \in R^{C'}$.

Finally, after obtaining the probability vector, attention mechanism is used to add the feature graphs according to the weights to realize the adaptive selection of different scale information by neurons.

The steps are detailed as follows. First, the softmax function is used to obtain the cth channel weight value

$$W_c^i = \frac{\exp V_c^i}{\sum_{j=1}^{4} \exp V_c^j} \tag{10}$$

where $W_c^i (i \in [1, 4])$ represents the weight value of the output feature map of the ith branch of the cth channel and $V_c^i (i \in [1, 4])$ represents the probability vector of the output feature map of the ith branch of the cth channel.

Second, the feature map $R^i (i \in 1\text{–}4)$ obtained from different branches of the convolutional layer is weighted and summed with the obtained weight value to generate the final output feature map F_{output}, where $F_{\text{output}} = [F_1, F_2, \ldots, F_c]$, and the output feature map of the cth channel is

$$F_{c_{\text{output}}} = \sum\nolimits_i {R_c}^i {W_c}^i \tag{11}$$

$$\sum\nolimits_i {W_c}^i = 1. \tag{12}$$

The above is the selective convolution module based on attention mechanism. The feature map is weighted and added according to the weight value, so that the receptive field of each layer of neurons is no longer a fixed size but adaptively adjusts the size of the receptive field according to the multi-scale information of the input information.

B. Design of Adaptive Receptive Field Enhancement Network Based on the Selective Convolution Module

1) VoVNetV2: VoVNetV2 [26] is a real-time and efficient feature extraction network for target detection that is composed of one-shot aggregation (OSA) module. The OSA module not only inherits the advantages of multireceptive field representation of multiple features of DenseNet [27] but also solves the problem of low efficiency caused by dense connections in DenseNet.

The OSA module is shown in Fig. 5. In the OSA, each convolutional layer consists of bidirectional joins: one joint is connected to the next layer to produce features with a larger receptive field, while the other joint is aggregated only into the final output feature map for feature aggregation. As shown in Fig. 5, the OSA module in the VoVNetV2 network adds the eSE module [28] on the last feature layer, and the eSE module further enhances features by using the FC layer.

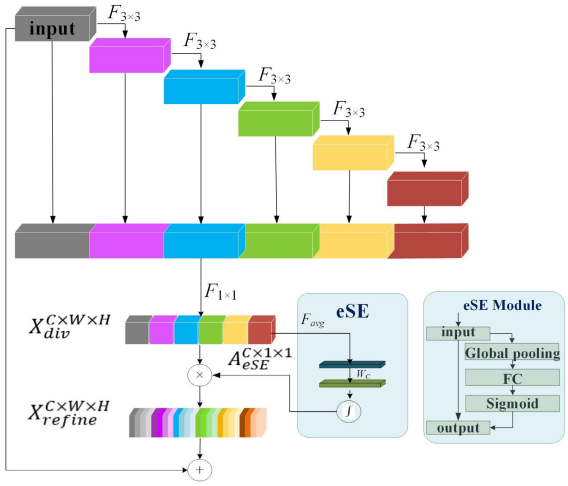


Fig. 5. OSA module.

Although the OSA module in the VoVNetV2 network is conducive to feature extraction, it directly aggregates the feature map at the last layer, which results in fixed neuron receptive field size at each layer. In addition, the OSA modules uses ordinary convolution in the branch and cannot achieve the maximum increase of the receptive field with a limited number of parameters. Therefore, to enable neurons to adjust the receptive field adaptively and enhance the feature map receptive field, improvements were made to the OSA module.

2) Adaptive Receptive Field Enhancement Network: Inspired by the OSA module in VoVNetV2, we propose an adaptive receptive field enhancement network based on attention mechanism in this article, and the overall design of the network is described as follows: first, after reducing the number of channels by 1×1 convolution, the input feature map is successively convolved by three K-convs with different expansion factors, and the outputs of these four convolutions are spliced over the channels to obtain a feature map with a very rich receptive field. Second, the feature map with rich receptive field obtains the channel weights through the selective convolutional module and operates the weights with the feature map to realize adaptive adjustment of the receptive field size. Third, the obtained feature map is recovered to the same number of channels as the input feature map by 1×1 convolution, and then, the feature map is further enhanced by the eSE module. Finally, the feature map is added to the original feature map by short-circuit connection. Fig. 6 shows the overall framework of the adaptive receptive field enhancement network.

Using the three series branches in Fig. 6 as an example, the specific description is presented as follows.

1) To reduce the amount of calculation, 1×1 convolution is used to change the number of channels to (1/4) of the input.
2) The input feature maps are successively passed through the series branches of the RFB convolutional layer, and the receptive field is enhanced through K-conv. As the transformation matrix F in K-conv has the identity submatrix of $r_2 \times r_2$, it can capture the local contextual

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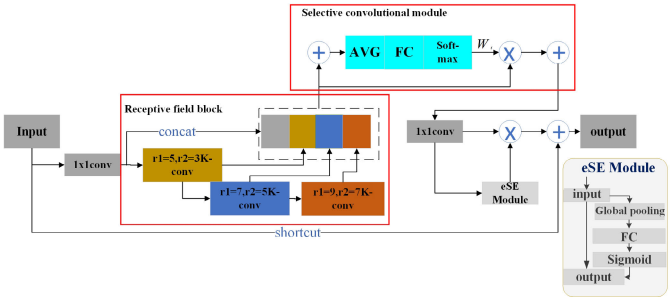


Fig. 6. Adaptive receptive field enhancement network.

information disregarded by dilated convolution. Thus, 3×3 convolution is not needed to refine the contextual information as in the RFB structure.

3) The feature maps output by series branches of the convolution layer are passed through the selective convolution module based on attention mechanism, so that the output feature graphs are weighted and added to the weight value rather than being simply concatenated, realizing the adaptive adjustment of the receptive field size according to the multiscale information of the input information.
4) The feature map is restored to the number of channels at the time of input by 1×1 convolution.
5) The eSE module of OSA module in VoVNetV2 was employed to further enhance the feature map, and a short-circuit connection was utilized to add the original feature map to obtain the final output feature map.

After adding the K-conv and series branches, the scale of the receptive field of the feature map is calculated as follows:

$$k' = (r_1 - 1) \times (k - 1) + k + r_2 - 1 \tag{13}$$

$$RF = RF_1 + k' - 1 \tag{14}$$

where r_1 represents an intermediate expansion factor, r_2 represents an internal sharing factor, k represents the size of the convolution kernel, k' represents the equivalent convolution kernel size after K-conv, RF_1 represents the receptive field size of the input feature map, and RF represent the receptive field size obtained by the K-conv.

Comparing (13) with (2) shows that the equivalent convolution kernel sizes of the dilated convolution and the K-conv are equal when $r_2 = 1$, and the equivalent convolution kernel size of the K-conv is larger than that of the dilated convolution when $r_2 > 1$. That is to say, the receptive field obtained by the proposed network in this article is larger than that of the RFB structure.

As can be seen from Table I, compared with the RFB structure and OSA module, with the same number of branches, the network proposed in this article has advantages in expanding the receptive field, capturing local contextual information, utilizing the convolutional resources, adjusting receptive field adaptively, and which is analyzed as follows.

First, both the dilated convolution in the RFB structure and the K-conv in our network are able to increase the size of the equivalent convolution kernel, which leads to an increase in the receptive field, whereas the ordinary convolution in the OSA module cannot. Second, since K-conv contains an intrasharing factor r_2 that is not present in dilated convolution, the network proposed in this article can fully utilize some of the local contextual information. Third, since the RFB structure is connected in parallel and both the OSA and our network are connected serially, the serial connection can fully utilize the convolutional resources to extract features and reduce the number of parameters of the network. Finally, both the RFB structure and the OSA module simply fuse the obtained receptive fields of different sizes, while our method can adaptively adjust the weights of the receptive fields according to the feature information of different layers.

IV. Experimental Validation and Results Analysis

A. Experimental Operating Environment

To verify the effectiveness of the proposed algorithm, Faster R-CNN (ResNet50 as the backbone network), YOLOv5 (YOLOv5-s model), and YOLOx (YOLOx-s) are selected as the basic algorithms, and the experimental testing environments are detailed as follows: Ubuntu20.04 OS, CPU Intel(R) Core(TM) i9-10940X @ 3.30 GHz, 48.0-GB RAM, NVIDA GeForce GTX 2080 Ti, and 11-GB video memory. The specific experimental parameters are shown in Table II.

B. Experimental Dataset

Nonsmall target categories, such as playgrounds, overpasses, ground track field, and so on, exist in all current publicly available aerial photography datasets. To specialize in object detection algorithms for aerial photography small targets, in this article, airplane, ship, and vehicle targets were selected from remote sensing object detection dataset (RSOD) [29] of Wuhan University, UCAS high resolution aerial object detection dataset (UCAS-AOD) [30] of the University of Science and Technology of China, and Northwestern Polytechnical University very high resolution-10 dataset (NWPU VHR-10) [31] of Northwestern Polytechnical University to construct an aerial small target dataset. Relevant experiments are carried out on a homemade aerial small target dataset. In this dataset, although the target pixels are small, there is minimal interference with the detection due to background occlusion of the target, which is suitable for analyzing the detection difficulties of insufficient feature information due to small targets in aerial images. According to the Microsoft Common Objects in Context (MS COCO) dataset [32], pixels smaller than 32×32 are classified as small targets, pixels larger than 96×96 are classified as large targets, and the remaining pixels are medium targets. Details of the aerial image small target dataset are shown in Table III.

Fig. 7 shows a selection of sample pictures from the dataset, airplanes in Fig. 7(a), ships in Fig. 7(b), and all vehicles in Fig. 7(c). The area of airplanes, ships, and cars in the picture is relatively small, among which the area of cars is the smallest compared with the other two types of targets. Moreover, three types of small targets are clustered in high density in the aerial images, which hinders object detection.

C. Evaluation Indicators

In this article, the mAP is used to detect the object detection quality, frames/s (FPS) is used to detect the detection speed of

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TABLE I
COMPARISON OF THE ADVANTAGES OF RFB, OSA, AND THE NETWORK PROPOSED IN THIS ARTICLE

Method	Increase the receptive field more	Capture of local contextual information	Utilizing convolutional resources effectively	Adaptive receptive field Enhancement
RFB structure	✔	✗	✗	✗
OSA module	✗	✔	✔	✔
ours	✔	✔	✔	✔



(a) (b) (c)

Fig. 7. Sample from the aerial small target dataset. (a) Sample airplane diagram. (b) Sample ship diagram. (c) Sample vehicle diagram.

TABLE II
EXPERIMENTAL PARAMETER DESIGN

Method	Faster R-CNN	YOLOx-s	YOLOv5-s
Epochs	300	300	300
Resolution	(640,640)	(640,640)	(640,640)
Number of GPU	2	2	2
Batch_Size	8	8	8
Depth	-	0.33	-
Width	-	0.5	-
Momentum	0.9	0.9	0.9
Learning Rate	0.02	0.05	0.01

TABLE III
AERIAL IMAGE SMALL TARGET DATASET

Classes	Number	Image scale	Scale
Training images	9158	Small target ratio	63%
Test images	1018	Medium target ratio	33.3%
Targets	119905	Large target ratio	3.7%
Plane samples	16235	Average area of plane	0.98%
Ship samples	62702	Average area of ship	0.41%
Vehicle samples	40968	Average area of vehicle	0.14%

each photograph, and parameters and giga floating-point operations per second (GFLOPS) are used to measure the model's size and computational requirements, respectively. The mAP metric consists of precision, recall, and average precision (AP)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

where TP represents that the prediction and actual are positive, FP represents that the prediction is positive and the actual is negative, TN represents that the prediction and actual are negative, and FN represents that the prediction is negative and the actual is positive.

The AP is the area delineated by the precision–recall (P–R) curve and the coordinate axis, which is used to measure the detection quality of a class of targets. The calculation formula is expressed as follows:

$$AP = \int_0^1 P(R)d(R). \tag{17}$$

When detecting multicategory targets, the mAP, which is used to evaluate the detection accuracy of multicategory targets, can be obtained by averaging the AP of all categories. The calculation formula is presented as follows:

$$\text{mAP} = \frac{1}{M}\sum_{i=1}^{\text{classes}} \int_0^1 P(R)d(R) \tag{18}$$

where M represents the number of categories $i \in (1, M)$.

D. Comparative Experiment Evaluation Analysis

1) Interval Expansion Factor, Internal Shared Factor Impact Test:

a) Objective evaluation: Since the setting of expansion rate and in K-conv affects the detection performance of adaptive receptive field enhancement network based on attention mechanism on small targets in aerial images, experiments are designed to select the expansion factor that will yield the best detection effect for the algorithm.

The literature [21] concluded that the interval expansion factor r_1 and the internal sharing factor r_2 are effective in the network: 1) as the internal sharing factor r_2 increases ($r_2 < r_1$), convolution can capture more contextual information; 2) increasing of the interval expansion factor r_1 sometimes leads to a larger receptive field on the feature map and improves the detection results; and 3) compared with the dilated convolution, K-conv can obtain better experimental results. Based on the above conclusions, the values of r_1 and r_2 are set, and the YOLOx-s algorithm is selected for comparative

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experiments. The expansion factors of RFBs in the adaptive receptive field enhancement network with the highest objective mAP index evaluation and the best subjective actual detection effect are selected. To make the feature map have more detailed information about target, the adaptive receptive field enhancement network based on attention mechanism is added to the neck layer of YOLOx for feature fusion to increase the receptive field of the feature map after feature fusion. In addition, since YOLOx is a multiscale object detection algorithm (detecting objects of different scales on different scale feature maps), adaptive field enhancement networks with different expansion rates r_1 and r_2 are set on the output layers of different scale feature maps, as shown in Fig. 8, where p_2, p_1, and p_0 represent output feature maps with decreasing sizes. The YOLOx detects small targets on p_2, medium targets on p_1, and large targets on p_0. The detection performance of the YOLOx algorithm after adding adaptive receptive field enhancement network modules with different expansion rates, namely, the evaluation value pair of mAP index, is shown in Table IV.

In view of the characteristics of YOLOx multiscale target detection, adaptive receptive field enhancement network modules with different expansion rates were added to the output layer of detection feature maps of small, medium, and large targets for experiment. The specific design is shown in Table IV. First, the experiments are conducted on receptive blocks designed with an increasing expansion factor (Nos. 2–4). Second, the experiment is conducted on receptive blocks designed with a decreasing expansion factor (No. 5). Experiments are conducted on receptive blocks designed with the same expansion rate (Nos. 6–8).

As shown in Table IV, compared with the basic algorithm of YOLOx (No. 1), the performance of the algorithm with the receptive field enhancement module has been improved. Further analysis shows that in the test experiments of Nos. 2–4, No. 3 has the best detection results, mainly as with an increase in the expansion factor, the receptive field increases, and the accuracy of target detection increases (such as Nos. 2 and 3). However, with the continuous increase in the expansion factor, the information of small and medium targets will be missing, which will reduce the detection accuracy (such as Nos. 3 and 4).

As shown in Table IV, the detection result of No. 5 is worse than that of No. 3, mainly as No. 5 adds a receptive field enhancement module with a large expansion factor to the feature map output layer, which is used to detect small targets and will cause missing information of small targets and medium targets. Therefore, the detection accuracy of medium and small objects decreases.

Experiment Nos. 6 and 7 keep the receptive field enhancement modules of two different feature map output layers unchanged, and the detection result is improved with an increase in the expansion factor. However, the detection result of No. 3 is still the best, as it can achieve superior detection results when detecting small, medium, and large objects. The experimental test of No. 8 keeps all receptive field modules the same, and its objective indicators are inferior to those of No. 3.

In summary, the receptive field module set according to No. 3 achieves the best detection results.

b) Subjective evaluation: The objective evaluation in Section IV-D1.a indicates that the object detection algorithm has a better performance after adding the adaptive receptive field enhancement network module set according to No. 3, but it is not intuitive enough. The following subjective evaluation uses the actual detection diagrams.

Fig. 9 shows the actual detection effect diagram of YOLOx-s with the adaptive receptive field enhancement network module with different expansion rates. Fig. 9(a) is the original drawing, and Fig. 9(b)–(i), respectively, corresponds to the actual test renderings of Nos. 1–8, respectively, in Table IV. According to the actual detection effect drawing, the detection diagram of No. 3 is the best. Fig. 9(b) corresponds to No. 1. There are two omissions in the detection graph of the original algorithm (points indicated by the yellow arrow). Fig. 9(c)–(e) corresponds to the detection diagrams of serial numbers 2–4, respectively, with the adaptive receptive field enhancement network module and an increased expansion factor. The detection diagrams of Nos. 2 and 4 have missed detections, while the detection diagrams of No. 3 do not. The main reason is that the excessive increase in the expansion factor will not increase the detection accuracy but will easily lead to a loss of medium and small targets. Thus, expansion factor can only be moderately increased. Fig. 9(f)–(i) corresponds to Nos. 5–8 of the actual detection diagrams, and the phenomenon of missing detection also appears. Therefore, combining the objective and subjective evaluation results, YOLOx with the adaptive receptive field enhancement network module set according to No. 3 yields the experimental results with the highest objective mAP index evaluation and the best subjective actual detection result.

2) Receptive Field-Enhanced Structural Performance Testing:

a) Objective evaluation: To verify the superiority of the adaptive receptive field enhancement network module compared with other receptive field enhancement modules in improving the detection accuracy and detection speed of the algorithm for small targets in aerial images, the YOLOx-s algorithm serves as the basis. Atrous spatial pyramid pooling (ASPP) [33], RFB [14], RFB-s [14], and the adaptive receptive field enhancement network module were added in the same position as Fig. 8. The adaptive receptive field enhancement network module was set according to No. 3 in Table IV and marked with *. Model training and testing were carried out on YOLOx-s algorithms added with different receptive field enhancement modules, and the detection performance of the algorithms added with several receptive field enhancement modules was compared by the mAP, FPS, Params, and GFLOPs, as shown in Table V.

It is shown in Table V that the detection performance of the algorithm is improved with the receptive field enhancement module, and that with the adaptive receptive field enhancement network module is the highest. Further analysis reveals that for the original algorithm with the ASPP module, the value

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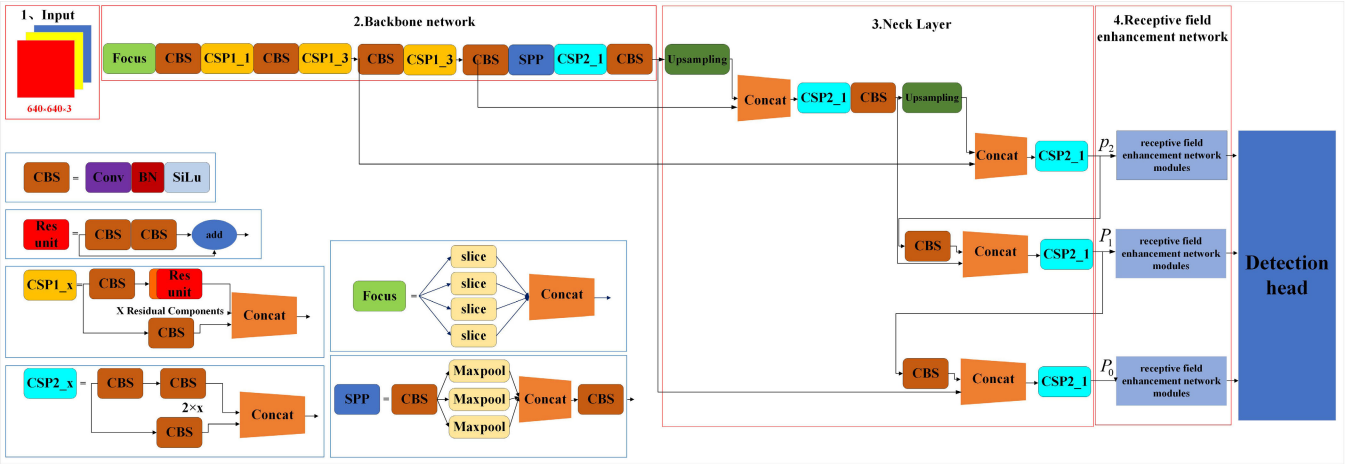


Fig. 8. YOLOx structure after adding the adaptive receptive field enhancement network.

TABLE IV

COMPARISON OF DETECTION PERFORMANCE AFTER ADDING ADAPTIVE RECEPTIVE FIELD ENHANCEMENT NETWORK MODULES WITH DIFFERENT EXPANSION RATES IN YOLOX

serial number	p_2	p_1	p_0	mAP_S	mAP_M	mAP_L	mAP_{50}	$mAP_{50:95}$
1	-	-	-	0.514	0.757	0.902	0.930	0.640
2	$r_1 = [2, 3, 5]$ $r_2 = [1, 1, 3]$	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	0.517	0.761	0.911	0.933	0.645
3	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	$r_1 = [7, 8, 11]$ $r_2 = [5, 7, 9]$	**0.542**	**0.770**	**0.933**	**0.945**	**0.661**
4	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	$r_1 = [7, 8, 11]$ $r_2 = [5, 7, 9]$	$r_1 = [9, 11, 13]$ $r_2 = [7, 9, 11]$	0.525	0.766	0.926	0.938	0.650
5	$r_1 = [7, 8, 11]$ $r_2 = [5, 7, 9]$	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	0.512	0.760	0.920	0.934	0.646
6	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	$r_1 = [5, 6, 9]$ $r_2 = [3, 5, 7]$	0.523	0.761	0.878	0.934	0.645
7	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [7, 8, 11]$ $r_2 = [5, 7, 9]$	$r_1 = [7, 8, 11]$ $r_2 = [5, 7, 9]$	0.521	0.764	0.903	0.937	0.648
8	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	$r_1 = [3, 4, 6]$ $r_2 = [1, 3, 5]$	0.519	0.760	0.902	0.932	0.642

TABLE V

COMPARISON OF DETECTION PERFORMANCE OF YOLOX-S-ENHANCED MODULES WITH DIFFERENT RECEPTIVE FIELDS

Method	mAP_S	mAP_M	mAP_L	mAP_{50}	$mAP_{50:95}$	FPS	$Params$	$GFlops$
YOLOx-s	0.514	0.757	0.902	0.930	0.640	**75**	**8.9M**	**26.65**
YOLOx-s-ASPP	0.522	0.763	0.908	0.933	0.648	61	20.6M	47.50
YOLOx-s-RFB	0.518	0.761	0.911	0.934	0.644	61	10.7M	29.83
YOLOx-s-RFB-s	0.521	0.763	0.915	0.937	0.646	58	10.9M	30.28
YOLOx-s*	**0.542**	**0.770**	**0.933**	**0.945**	**0.661**	63	10.4M	28.74

of mAP_{50} has increased by 0.3%, and the value of $mAP_{50:95}$ has increased by 0.8%. For the original algorithm with the RFB module, the value of mAP_{50} has increased by 0.4%, and the value of $mAP_{50:95}$ has increased by 0.4%. For the original algorithm with the RFB-s module, the value of mAP_{50} has increased by 0.7%, and the value of $mAP_{50:95}$ increased by 0.6%. For the original algorithm with the module proposed in this article, the mAP_{50} has increased by 1.5%, and the value of $mAP_{50:90}$ increased by 2.1%, and among all the models, our method has the highest mAP_S, mAP_M, and mAP_L, indicating that our method has a great potential to improve the detection accuracy of small targets. For a more intuitive comparison, Fig. 10 shows 300 rounds of changes in the mAP value when YOLOx-s added different receptive field enhancement modules. As shown in Fig. 10, the surge occurred in the last 15 rounds due to the large amount of inaccurate labeling introduced by the mosaic data augmentation used by YOLOx in training, as well as the fact that mix-up alters the distribution of the training data, and switching off the data augmentation in the last 15 epochs allows the detector to complete its final convergence on the real-world data. Fig. 10 shows that the algorithm with the adaptive receptive field enhancement network module proposed in this article has the best detection performance.

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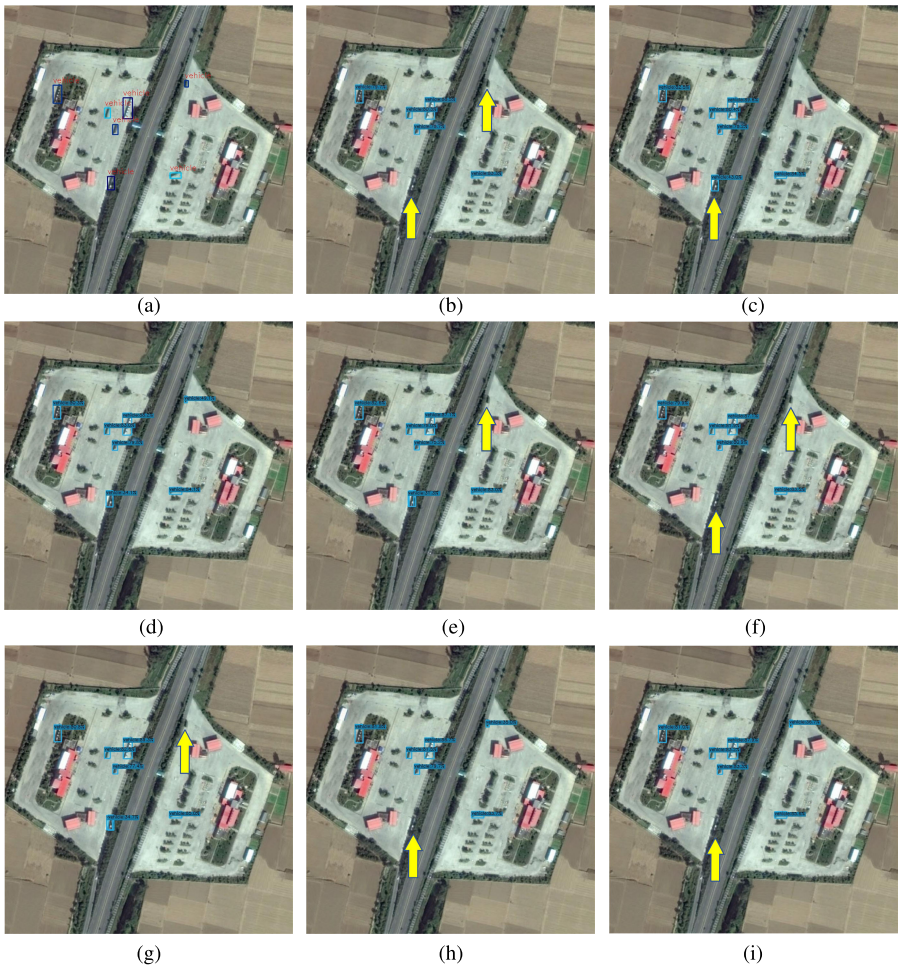
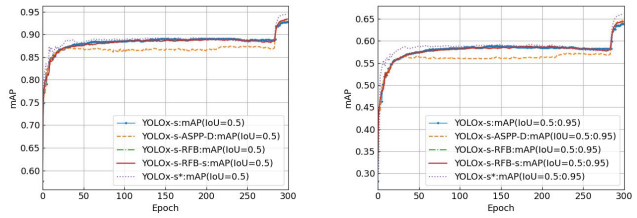


Fig. 9. Actual detection of the YOLOx-s adaptive receptive field enhancement network module with different expansion rates. (a) Original image. (b) No. 1. (c) No. 2. (d) No. 3. (e) No. 4. (f) No. 5. (g) No. 6. (h) No. 7. (i) No. 8.



(a) (b)

Fig. 10. Three-hundred rounds of changes in the mAP value in YOLOx-s with different receptive field enhancement modules. (a) mAP(IoU $=$ 0.5). (b) mAP(IoU $= 0.5 : 0.95$).

In addition, the FPS, parameters, and GFLOPs values of YOLOx-s with different receptive field enhancement modules were calculated to compare model efficiency. As shown in Table V, the FPS, Params, and GFLOPs values of the adaptive receptive field enhancement network module added by YOLOx-s are greater than other receptive field modules. Therefore, from the perspective of module efficiency, the algorithm added by the proposed module in this article has the fastest detection speed and minimal hardware requirements.

In summary, with regard to either the detection accuracy or detection speed, the improved algorithm with the addition of receptive field enhancement module is compared, and the receptive field enhancement module proposed in this article is superior to other receptive field enhancement modules.

b) Subjective evaluation: According to objective evaluation, the algorithm with the adaptive field enhancement network module proposed in this article has the best performance. However, the algorithm is not intuitive enough. The following subjective evaluation is based on the actual detection diagram.

Fig. 11 shows the actual detection diagram of YOLOx-s with different receptive field enhancement modules. Fig. 11(a) shows the original diagram, Fig. 11(b) shows the original algorithm detection diagram, Fig. 11(c) shows the detection diagram of YOLOx-s with the ASPP RFB, Fig. 11(d) shows the detection diagram of YOLOx-s with the RFB, Fig. 11(e) shows the detection diagram of YOLOx-s with the RFB-s receptive field block, and Fig. 11(f) shows the detection diagram of YOLOx-s with the adaptive receptive field enhancement network module proposed in this article. In Fig. 11, yellow arrows indicate the missed detections, and red arrows indicate false detections. As shown in Fig. 11, YOLOx-s with the receptive field enhancement module proposed in this article performs best for small targets in aerial images. Therefore, combining the objective and subjective

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Fig. 11. Actual detection of YOLOx-s with different receptive field structures. (a) and (g) Original image. (b) and (h) YOLOx-s. (c) and (i) YOLOx-s-ASPP. (d) and (j) YOLOx-s-RFB. (e) and (k) YOLOx-s-RFB-s. (f) and (l) YOLOx-s*.

evaluation results, the algorithm with the adaptive receptive field enhancement network module proposed in this article has the best detection performance for small targets in aerial images.

3) Adaptive Receptive Field Enhancement Network Module Effectiveness Test:

a) Objective evaluation: To test the effectiveness of adding an adaptive field enhancement network module to different target detection algorithms to improve the detection accuracy of small targets in aerial images, the module is added to Faster R-CNN, YOLOv5-s, and YOLOx-s, respectively. The test results are shown in Table VI, where * represents the algorithm after adding the module. Since both YOLOv5-s and YOLOx-s adopted multiscale feature prediction, according to the test results in Table IV, the neck layer of YOLOv5-s and YOLOx-s underwent feature fusion processing at the same position (Fig. 8) was followed by the adaptive receptive field enhancement network module set according to No. 3 in Table IV. The Faster R-CNN algorithm is followed by the backbone network with an adaptive receptive field enhancement network module with an expansion factor set to $r_1 = [3, 4, 6]$ and $r_2 = [1, 3, 5]$.

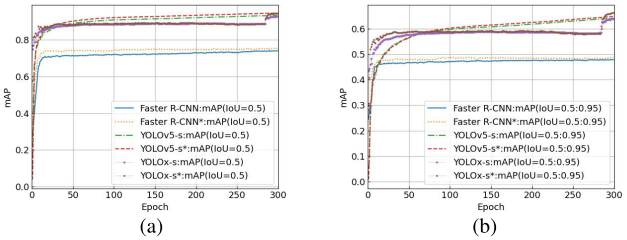
Table VI shows that the detection performance of aerial images is improved after the addition of the adaptive receptive field enhancement module. As shown in Table VI, for Faster R-CNN with the module, the value of mAP_{50} has increased

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TABLE VI

COMPARISON BETWEEN THE ORIGINAL DETECTION PERFORMANCE OF DIFFERENT ALGORITHMS AND THE DETECTION PERFORMANCE AFTER THE ADDITION OF THE ADAPTIVE RECEPTIVE FIELD ENHANCEMENT NETWORK MODULE

Method	mAP_S	mAP_M	mAP_L	mAP_{50}	$mAP_{50:95}$
Faster R-CNN	0.304	0.657	0.761	0.740	0.478
Faster R-CNN*	0.314	0.664	0.758	0.755	0.487
YOLOv5-s	0.481	0.748	0.882	0.934	0.640
YOLOv5-s*	0.519	0.767	0.919	**0.946**	0.647
YOLOx-s	0.514	0.757	0.902	0.930	0.640
YOLOx-s*	**0.542**	**0.770**	**0.933**	0.945	**0.661**



(a) (b)

Fig. 12. Three-hundred rounds of changes in the mAP value in YOLOx-s with different receptive field enhancement modules. (a) mAP(IoU $=$ 0.5). (b) mAP(IoU $=$ 0.5 : 0.95).

TABLE VII

COMPARISON BETWEEN THE ORIGINAL DETECTION SPEED OF DIFFERENT ALGORITHMS AND THE DETECTION SPEED AFTER ADDING THE ADAPTIVE RECEPTIVE FIELD ENHANCEMENT NETWORK

Method	FPS
YOLOv5-s	**67**
YOLOv5-s*	50
YOLOx-s	**80**
YOLOx-s*	63

by 1.5%, and the value of $mAP_{50:95}$ has increased by 0.9%. For YOLOv5-s with the module, the value of mAP_{50} has increased by 1.2%, and the value of $mAP_{50:95}$ has increased by 0.7%. For YOLOx-s with the module, the value of mAP_{50} has increased by 1.5%, and the value of $mAP_{50:95}$ has increased by 2.1%. Furthermore, from the experimental results of mAP_S, mAP_M, and mAP_L, the addition of the adaptive receptive field enhancement network improves all the index values of YOLOx-s and YOLOv5-s as well as the mAP_S and mAP_M of Faster R-CNN, while the mAP_L of Faster R-CNN decreases slightly. The above analyses show that our method helps to improve the accuracy of object detection, especially for small- and medium-sized targets. To make a more intuitive comparison, Fig. 12 shows 300 rounds of mAP value changes for each algorithm before and after the addition of the adaptive receptive field enhancement network module. Fig. 12 shows that the detection algorithm after the addition of the adaptive receptive field enhancement network module has higher mAP values than the original algorithm.

To further compare the performance of the improved algorithm, FPS values were calculated to compare the detection speeds. As shown in Table VII. As the objective evaluation value of Faster R-CNN is lowest, only the FPS values of YOLOv5-s and YOLOx-s were calculated. According to Table VII, although the detection speed of the algorithm decreases after the addition of the receptive field enhancement network module, the decrease is minimal. Moreover, YOLOx-s still has a faster detection speed.

b) Subjective evaluation: The objective evaluation shows that the algorithm performance is improved after the addition of the adaptive receptive field enhancement network module. The following subjective evaluation is based on the actual detection diagram. Fig. 13 shows a comparison of actual detection before and after the improvement in the Faster R-CNN algorithm, Fig. 14 shows a comparison of actual detection before and after the improvement of the YOLOv5-s algorithm, and Fig. 15 shows a comparison of actual detection before and after the improvement in the YOLOx-s algorithm. The yellow arrow points to missed detection, and the red arrow points to false detections.

Fig. 13 shows four groups of comparison graphs of actual detection before and after the improvement in the Faster R-CNN algorithm. Fig. 13(a) corresponds to the original graph, Fig. 13(b) corresponds to the original algorithm detection graph, and Fig. 13(c) corresponds to the detection graph after the algorithm is added to the adaptive receptive field enhancement network module. In Fig. 13(b), (e), and (h), the red arrows point to false detections. The point of the yellow arrow in Fig. 13(k) indicates that there are four missed detections by the original algorithm, while the point of the yellow arrow in Fig. 13(l) indicates that there are only two missed detections by the improved algorithm. In conclusion, Faster R-CNN with an adaptive receptive field enhancement network module has better detection performance on small targets in aerial images.

Fig. 14 shows four groups of comparison diagrams of actual detection before and after the improvement in the YOLOv5-s algorithm. Fig. 14(a) corresponds to the original diagrams, Fig. 14(b) corresponds to the original algorithm detection diagrams, and Fig. 14(c) corresponds to the detection diagrams of the algorithm with the adaptive receptive field enhancement network module. In Fig. 14(b) and (e), there are false detections at the point indicated by the red arrow, and in Fig. 14(h) and (k), there are missed detections at the point indicated by the yellow arrow. In particular, the missing target in Fig. 14(k) is successfully detected in Fig. 14(i), even though it is not marked in Fig. 14(j). In conclusion, YOLOv5-s with

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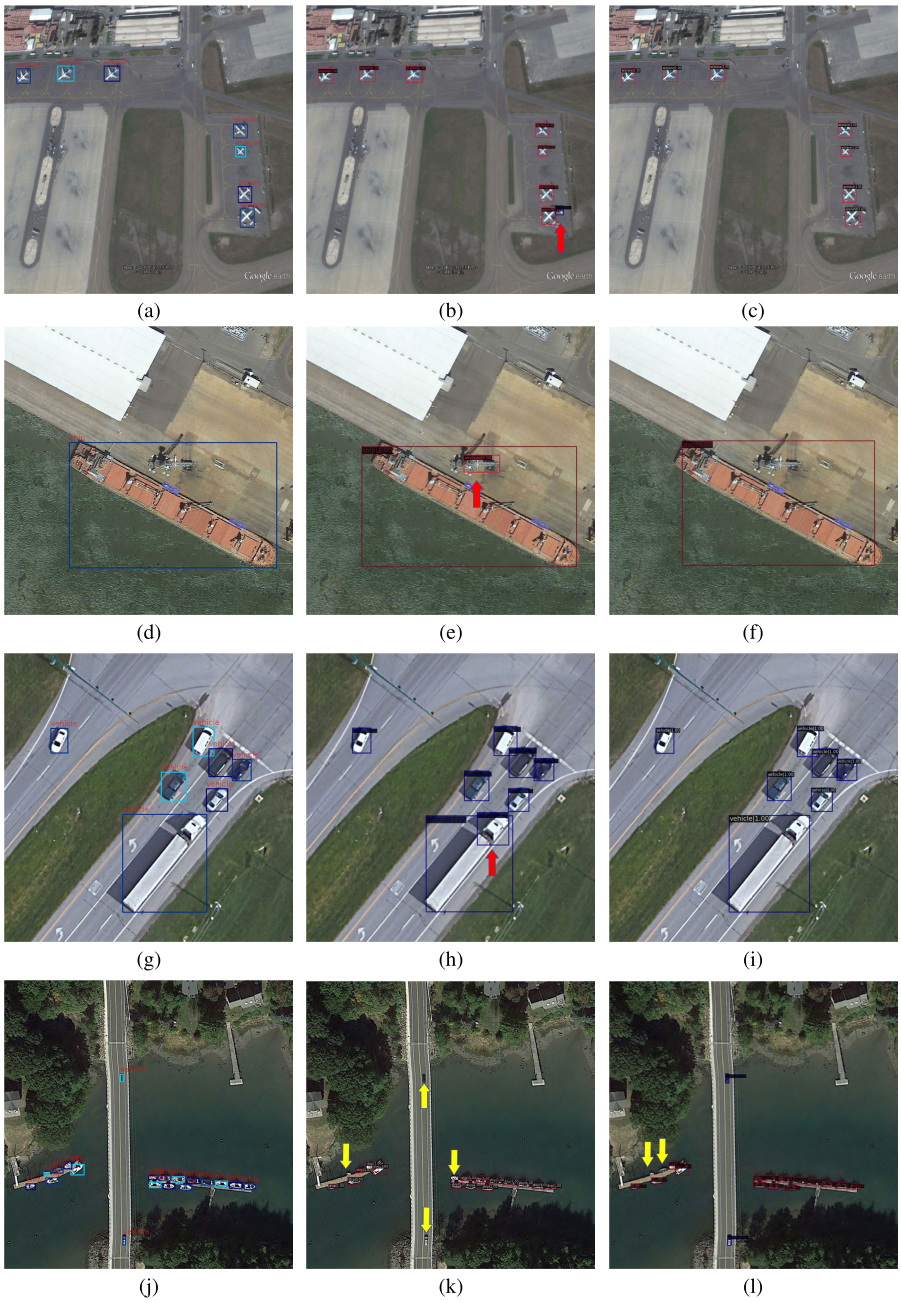


Fig. 13. Comparison of actual detection before and after the improvement in the Faster R-CNN algorithm. (a), (d), (g), and (j) Original image. (b), (e), (h), and (k) Original method. (c), (f), (i), and (l) Method improvement.

TABLE VIII
COMPARISON RESULTS OF DIFFERENT ALGORITHMS ON HOMEMADE SMALL TARGET DATASETS

Method	mAP_S	mAP_M	mAP_L	mAP_{50}	$mAP_{50:95}$	$Params$	$GFlops$
DETR	0.253	0.630	0.820	0.744	0.447	41.2M	86.3
YOLOv5-s	0.481	0.748	0.882	0.934	0.640	7.2M	15.8
YOLOv7-tiny	0.429	0.739	0.843	0.905	0.603	**6.1**M	**13.3**
YOLOx-s	0.514	0.757	0.902	0.930	0.640	8.9M	26.7
YOLOv8-s	0.518	0.761	0.899	0.931	0.652	11.2M	28.5
GLSAN	0.471	0.709	0.853	0.918	0.641	-	-
Ours	**0.542**	**0.770**	**0.933**	**0.945**	**0.661**	10.4M	28.7

an adaptive receptive field enhancement network module has better detection performance on small targets in aerial images.

Fig. 15 shows the comparison diagram of actual detection before and after the improvement in the YOLOx-s

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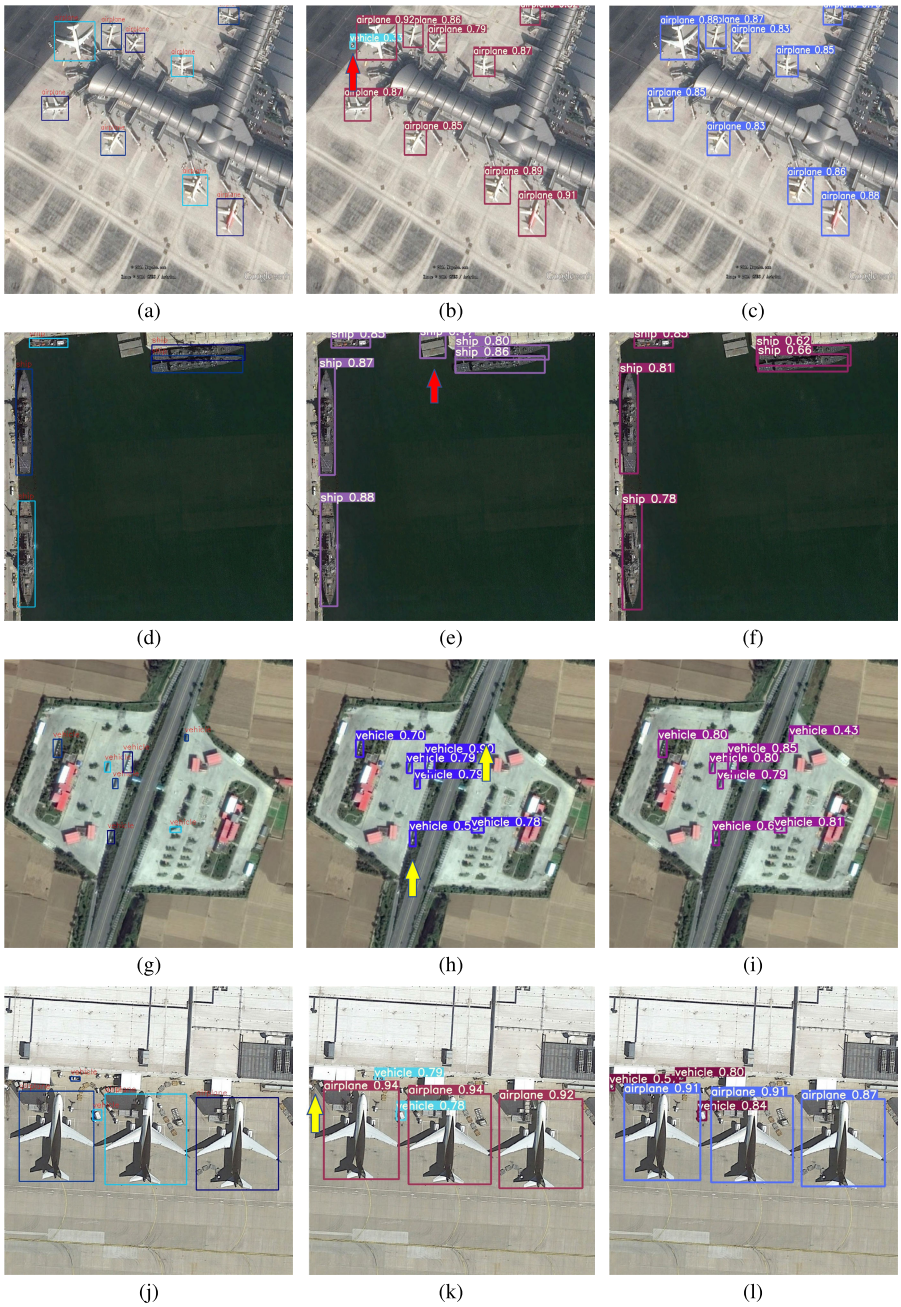


Fig. 14. Comparison of actual detection before and after the improvement in the YOLOv5-s algorithm. (a), (d), (g), and (j) Original image. (b), (e), (h), and (k) Original method. (c), (f), (i), and (l) Method improvement.

TABLE IX
COMPARISON RESULTS OF DIFFERENT ALGORITHMS ON VISDRONE2021

Method	mAP_S	mAP_M	mAP_L	mAP_{50}	$mAP_{50:95}$	$Params$	$GFlops$
YOLOv5-s	0.094	0.250	0.359	0.314	0.172	14.4M	15.8
YOLOv7-tiny	0.102	0.246	0.344	0.330	0.168	12.3M	**13.3**
YOLOx-s	0.117	0.297	0.434	0.364	0.205	**8.9**M	26.7
YOLOv8-s	0.116	0.308	0.404	0.373	0.223	22.5M	28.5
Ours	**0.132**	**0.314**	**0.453**	**0.390**	**0.229**	10.4M	28.7

algorithm. Fig. 15(a) corresponds to the original diagram, Fig. 15(b) corresponds to the original algorithm detection diagram, and Fig. 15(c) corresponds to the detection diagram of the algorithm with the adaptive receptive field enhancement network module. The point indicated by the red arrow in Fig. 15(b) and (e) indicates that the algorithm misdetected

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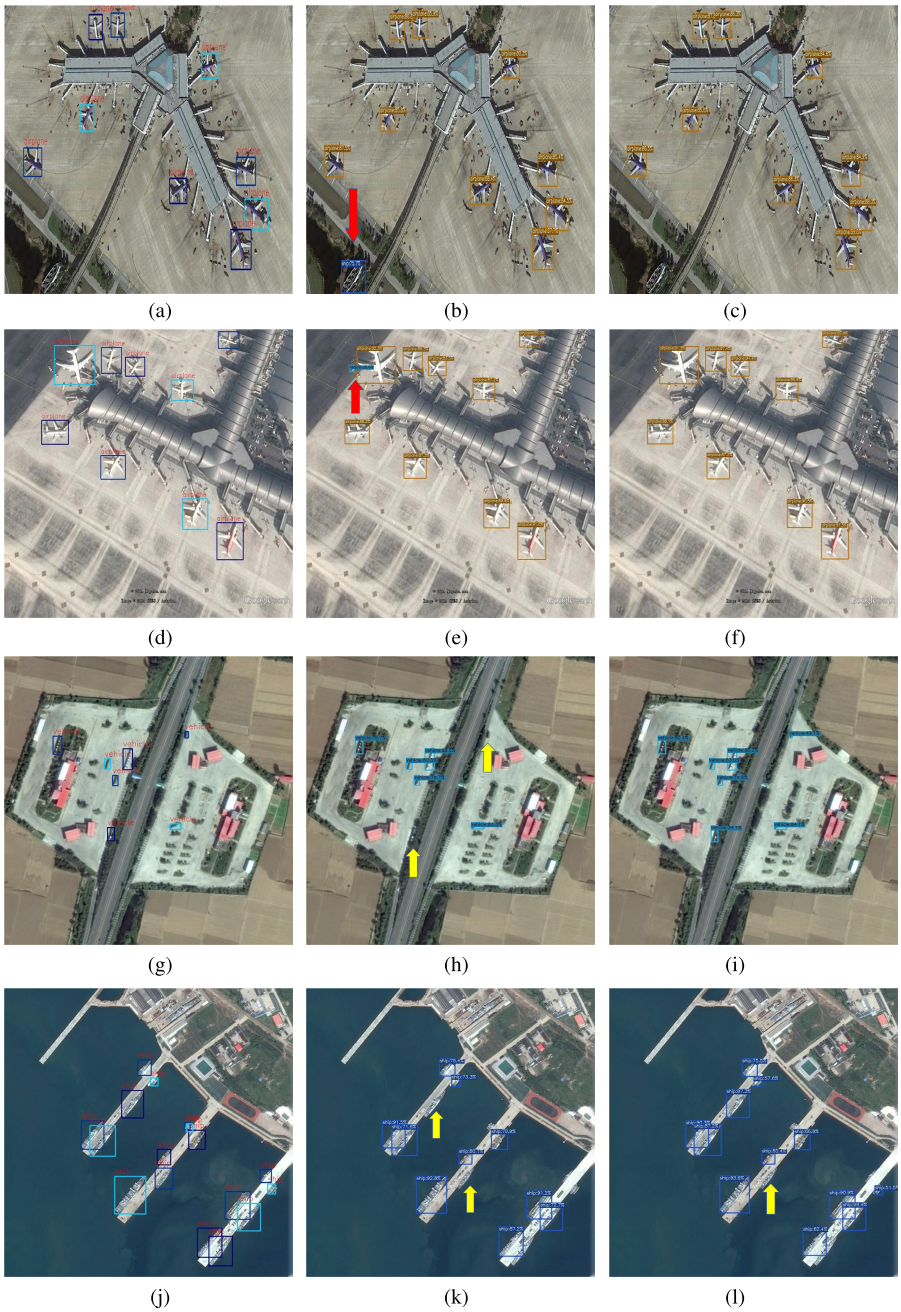


Fig. 15. Comparison of actual detection before and after the improvement in the YOLOx-s algorithm. (a), (d), (g), and (j) Original image. (b), (e), (h), and (k) Original method. (c), (f), (i), and (l) Method improvement.

nontarget objects as target objects by the original algorithm. In Fig. 15(h) and (k), there are missed detections at the point indicated by the yellow arrow. In conclusion, YOLOx-s with the adaptive receptive field enhancement network module has better detection performance on small targets in aerial images.

In conclusion, this section verifies the comparison of actual detection graphs before and after adding three different algorithms to the adaptive receptive field enhancement network that was added to three different algorithms. According to the actual test results, most of the algorithms with adaptive-enhanced receptive field network can effectively improve the false detection and missing detection of small targets in aerial images.

Combining the subjective and objective evaluation results, the adaptive field enhancement network can effectively improve the false detection and missing detection of small targets in aerial images by different target detection algorithms.

4) Comparison Results of Different Algorithms: We compare our network with different algorithms: 1) Transformer-based object detector detection transformer (DETR) [34]; 2) the YOLO family of object detectors includes YOLOv5 [18], YOLOv7 [35], YOLOx [16], and YOLOv8 [36]; and 3) image cropping algorithm GLSAN [3] with YOLOx-s as object detector. The parameter settings are kept the same, and the obtained experimental results are shown in Table VIII.

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Fig. 16. Detection examples of our method.

As can be seen from Table VIII, our method exceeds the other algorithms by a large margin and is optimal in all AP metrics while keeping parameters and GFLOPs at a low level.

We also evaluate our network on VisDrone2021. As reported in Table IX, our method outperforms other algorithms and achieves a gain of 2.4% in $mAP_{50:90}$ and 1.5% in mAP_S, compared with the baseline network with an increase of 1.5M parameters and 2 GFLOPs.

Fig. 16 illustrates the detection examples in the aerial small target dataset by our method; for better visualization, we do not show the class label.

V. Conclusion

To address the problem of insufficient semantic feature information caused by small objects in aerial images, this article proposes an adaptive receptive field enhancement network based on attention mechanism. The network can adjust the receptive field size of neurons at each layer, solve the problem that local contextual information cannot be captured when the receptive field is enlarged by the dilated convolution method, and with the same number of branches in the RFB, obtain receptive fields with a larger scale than the RFB structure, and effectively utilize the convolutional resources to reduce the computational effort. In this article, the performance of the adaptive receptive field enhancement network is tested and verified on a homemade small target aerial photography dataset. On the homemade, small target aerial photography dataset, first, based on the YOLOx algorithm, adaptive receptive field enhancement network modules with different expansion factors were set up. The adaptive receptive field enhancement network modules with the highest objective evaluation of the mAP value and the best subjective evaluation of the actual detection diagram were selected through experiments. Second, based on the YOLOx algorithm, a variety of other receptive field enhancement networks were added, and the comprehensive subjective and objective evaluation verified the superiority of the proposed network in improving the accuracy of the algorithm in detecting small targets in aerial images. Third, the adaptive receptive field enhancement network proposed in this article was added to Faster R-CNN and YOLOv5, and the comprehensive subjective and objective evaluation further verified that different object detection algorithms with the adaptive field enhancement network can effectively reduce the false detection and missing detection of small targets in aerial images. Finally, we compare our method with other algorithms on homemade datasets and VisDrone2021 to verify the superiority of the network proposed in this article. To further improve the performance of the detector, reconstructing the

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backbone using the network structure proposed in this article to obtain a network with robust feature extraction for aerial small target images is a compelling option that we will explore in future work.

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Jian Wang (Member, IEEE) received the B.E. degree in communication countermeasure from the National Defense Science and Technology, University of Electronic Countermeasure Technology, Hefei, China, in 1994, the M.S. degree in signal and information processing from the Aeronautical Engineering Institute, Air Force Engineering University, Xi'an, China, in 2000, and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, in 2005.

From 2010 to 2023, he was an Associate Professor at the School of Electronics and Information, Northwestern Polytechnical University, where he is working at the No. 365 Institute for the unmanned aerial vehicle (UAV) system design. His research interests include UAV intelligent processing technology, UAV ground observation video signal processing technology, and multisource information intelligent processing technology.

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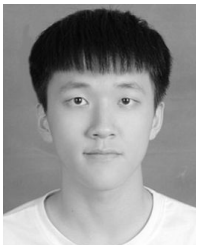
Xinqi Li was born in 2000. He received the B.E. degree from the School of Information Engineering, Nanchang University, Nanchang, China, in 2022. He is currently pursuing the master's degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include machine learning and deep learning.



Lihui Zhou was born in 1998. He received the B.E. degree from the College of Electronics and Information, Qingdao University, Qingdao, China, in 2022. He is currently pursuing the master's degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include multisource data fusion, image fusion, and machine learning.



Jiafu Chen was born in 2000. He received the B.E. degree from the School of Electronic and Information Engineering, Fujian Agriculture and Forestry University, Fuzhou, China, in 2022. He is currently pursuing the master's degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include image fusion, machine learning, and deep learning.



Zihao He was born in 2000. He received the B.E. degree from the School of Physical Science and Technology, Hebei University, Baoding, China, in 2021. He is currently pursuing the master's degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include multisource information processing, deep learning, and embedded development.



Linyang Guo was born in 2000. He received the B.E. degree from the School of Electronics and Information Engineering, Northwestern Polytechnical University, Xi'an, China, in 2022, where he is currently pursuing the master's degree.

His research interests include field-programmable gate array (FPGA) and signal processing.



Jie Liu was born in 1997. She received the B.E. degree from the School of Electronics and Information Engineering, Nanjing University of Posts and Telecommunications, Nanjing, China, in 2020. She is currently pursuing the master's degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

Her research interests include image fusion, machine learning, and deep learning.

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